

SIMATS ENGINEERING

ASSESSMENT TOOL 3

MCQ Test

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

C03: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 20%

QUESTIONS:20

Total Marks:20

1. Which TCP mechanism reduces the congestion window by half upon detecting packet loss via duplicate ACKs?
 - a) Slow Start
 - b) Fast Retransmit
 - c) Fast Recovery
 - d) Tahoe Reset(BL4)(1)
2. A TCP sender has CWND = 8 MSS and SSTHRESH = 16. After one RTT with no loss, CWND becomes:
 - a) 9 MSS
 - b) 16 MSS
 - c) 12 MSS
 - d) 8 MSS(BL4)(1)
3. Which field in the TCP header is used to implement flow control at the receiver side?
 - a) Sequence number
 - b) Acknowledgement number
 - c) Window size
 - d) Urgent pointer(BL4)(1)
4. Nagle's algorithm causes increased latency primarily because it:
 - a) Disables ACK piggybacking
 - b) Buffers small segments until ACK arrives
 - c) Reduces CWND on every packet

- d) Increases SYN backlog
(BL4)(1)
- 5. In a high-latency satellite link (600 ms RTT), TCP throughput is primarily limited by:
 - a) MTU size
 - b) CWND growth rate relative to RTT
 - c) DNS resolution time
 - d) UDP competition(BL4)(1)
- 6. UDP is preferred over TCP for DNS queries primarily because:
 - a) UDP supports fragmentation
 - b) DNS responses are small and connection overhead is unnecessary
 - c) TCP cannot use port 53
 - d) UDP has better error correction(BL4)(1)
- 7. A UDP-based streaming application experiences 15% packet loss. The best application-layer mitigation is:
 - a) Increase UDP buffer size
 - b) Switch to TCP
 - c) Implement Forward Error Correction (FEC)
 - d) Reduce frame rate to 1 fps(BL4)(1)
- 8. What is the maximum UDP payload size without IP-level fragmentation on a standard Ethernet link (MTU = 1500 bytes)?
 - a) 1480 bytes
 - b) 1472 bytes
 - c) 1460 bytes
 - d) 1500 bytes(BL4)(1)
- 9. QUIC protocol uses UDP as its transport. The key advantage over TCP-based TLS is:
 - a) Eliminating IP header overhead
 - b) 0-RTT or 1-RTT connection setup
 - c) Removing encryption
 - d) Native multicast support(BL4)(1)
- 10. HTTP/1.1 persistent connections reduce latency by:
 - a) Enabling UDP fallback
 - b) Reusing the same TCP connection for multiple requests
 - c) Compressing HTTP headers
 - d) Sending requests without waiting for DNS(BL4)(1)
- 11. Head-of-line blocking in HTTP/1.1 occurs when:
 - a) The DNS server is slow

- b) A response blocks subsequent responses on the same connection
 - c) TCP CWND drops to zero
 - d) The server rejects pipelined requests(BL4)(1)
- 12. HTTP/2 multiplexing solves HOL blocking at the HTTP layer but not at the:
 - a) Application layer
 - b) DNS layer
 - c) TCP layer
 - d) IP layer(BL4)(1)
- 13. Which HTTP status code indicates that a resource has permanently moved and browsers should update bookmarks?
 - a) 301
 - b) 302
 - c) 304
 - d) 307(BL4)(1)
- 14. In HTTP caching, the Cache-Control: no-cache directive means:
 - a) Never cache this resource
 - b) Validate with the server before using a cached copy
 - c) Cache for 0 seconds
 - d) Disable all caching headers(BL4)(1)
- 15. Which DNS record type maps a domain name to an IPv6 address?
 - a) A
 - b) CNAME
 - c) AAAA
 - d) MX(BL4)(1)
- 16. A DNS resolver switches from UDP to TCP when:
 - a) The query type is MX
 - b) The response is truncated (TC bit set)
 - c) TTL exceeds 3600 s
 - d) The authoritative server is unreachable(BL4)(1)
- 17. Which attack exploits the UDP-based DNS protocol by sending forged responses with a matching transaction ID?
 - a) ARP poisoning
 - b) DNS cache poisoning
 - c) SYN flooding
 - d) BGP hijacking(BL4)(1)
- 18. Setting a very low TTL (e.g., 30 s) for a DNS record primarily increases:
 - a) Cache hit ratio

- b) Recursive resolver query load
 - c) DNSSEC signature validity
 - d) Zone transfer frequency
- (BL4)(1)
19. A user browses to a new website. The correct sequence of protocol interactions for the first byte of the page is:
- a) HTTP → TCP → DNS
 - b) DNS → TCP handshake → HTTP GET
 - c) TCP → DNS → HTTP
 - d) DNS → HTTP → TCP
- (BL4)(1)
20. Under 20% packet loss, which protocol combination gives the best performance for a real-time multiplayer game?
- a) TCP for game state, HTTP for assets
 - b) UDP with application-layer sequencing for game state, HTTP/3 for assets
 - c) HTTP/2 for all traffic
 - d) TCP with Nagle enabled for all traffic
- (BL4)(1)

Code of Conduct:

*I **MOHAMED NAWAS N(192521 037)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: MOHAMED NAWAS N

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts

		problem-based questions			
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

Answers with Justification

1. Which TCP mechanism reduces the congestion window by half upon detecting packet loss via duplicate ACKs?

Answer: c) Fast Recovery

Justification: Fast Recovery is triggered after TCP receives multiple duplicate ACKs, indicating packet loss. Instead of returning to the initial congestion window, TCP reduces the congestion window, typically to about half of its previous value, and continues transmission using congestion avoidance. This helps maintain better throughput than restarting with Slow Start.

2. A TCP sender has CWND = 8 MSS and SSTHRESH = 16. After one RTT with no loss, CWND becomes:

Answer: b) 16 MSS

Justification: Since CWND (8 MSS) is below SSTHRESH (16 MSS), TCP is operating in the **Slow Start** phase. During Slow Start, the congestion window approximately doubles every RTT. Therefore, after one RTT:

$CWND = 8 \times 2 = 16 \text{ MSS}$

3. Which field in the TCP header is used to implement flow control at the receiver side?

Answer: c) Window size

Justification: The TCP **Window Size** field indicates how much additional data the receiver is currently able to accept. The sender uses this advertised receive window to control the amount of unacknowledged data it sends, preventing the receiver's buffer from becoming overwhelmed.

4. Nagle's algorithm causes increased latency primarily because it:

Answer: b) Buffers small segments until ACK arrives

Justification: Nagle's algorithm is designed to reduce network overhead by combining small TCP segments into larger ones. It may temporarily buffer small amounts of data until an acknowledgment is received or enough data is available. This waiting can increase latency, particularly in interactive applications that send many small messages.

5. In a high-latency satellite link (600 ms RTT), TCP throughput is primarily limited by:

Answer: b) CWND growth rate relative to RTT

Justification: A 600 ms RTT means that acknowledgments take a long time to return. TCP's congestion window grows based on successful transmissions and acknowledgments, so a large RTT makes congestion-window growth slower in real time. Consequently, TCP may take a long time to reach a window size sufficient to fully utilize a high-bandwidth satellite connection.

6. UDP is preferred over TCP for DNS queries primarily because:

Answer: b) DNS responses are small and connection overhead is unnecessary

Justification: Traditional DNS queries are usually small and can be handled efficiently using UDP without establishing a TCP connection. UDP avoids the TCP three-way handshake, reducing latency and protocol overhead. DNS can use TCP when required, such as when a response is too large or truncated.

7. A UDP-based streaming application experiences 15% packet loss. The best application-layer mitigation is:

Answer: c) Implement Forward Error Correction (FEC)

Justification: UDP does not automatically retransmit lost packets. **Forward Error Correction (FEC)** adds redundant information that allows the receiver to reconstruct some lost data without waiting for retransmission. This is particularly useful for real-time streaming, where retransmission may arrive too late.

8. What is the maximum UDP payload size without IP-level fragmentation on a standard Ethernet link (MTU = 1500 bytes)?

Answer: b) 1472 bytes

Justification: An Ethernet MTU of 1500 bytes includes the IP packet. For IPv4:

Ethernet MTU = 1500 bytes

IPv4 header = 20 bytes

UDP header = 8 bytes

Therefore:

UDP payload = 1500 – 20 – 8 = 1472 bytes

Thus, **1472 bytes** is the maximum UDP payload assuming a standard 20-byte IPv4 header and no IP options.

9. QUIC protocol uses UDP as its transport. The key advantage over TCP-based TLS is:

Answer: b) 0-RTT or 1-RTT connection setup

Justification: QUIC integrates transport and TLS security and can establish secure connections with fewer round trips than traditional TCP plus TLS. With a previously established connection, QUIC can support **0-RTT** data in appropriate circumstances, while a new connection can generally achieve secure establishment in about **1 RTT**. This reduces connection setup latency.

10. HTTP/1.1 persistent connections reduce latency by:

Answer: b) Reusing the same TCP connection for multiple requests

Justification: Persistent connections allow multiple HTTP requests and responses to use the same TCP connection. This avoids repeatedly performing the TCP three-way handshake, reducing connection establishment overhead and improving webpage loading performance.

11. Head-of-line blocking in HTTP/1.1 occurs when:

Answer: b) A response blocks subsequent responses on the same connection

Justification: In HTTP/1.1, requests and responses can be affected by ordering requirements on a connection. If an earlier response is delayed, subsequent responses can be prevented from being delivered or processed as intended. This is known as **Head-of-Line (HOL) blocking** and can increase application-level latency.

12. HTTP/2 multiplexing solves HOL blocking at the HTTP layer but not at the:

Answer: c) TCP layer

Justification: HTTP/2 allows multiple streams to share one TCP connection, reducing HTTP-level HOL blocking. However, TCP delivers bytes reliably and in order. If a TCP segment is lost, later data may have to wait until the missing segment is retransmitted. Therefore, HTTP/2 cannot eliminate **TCP-level HOL blocking**.

13. Which HTTP status code indicates that a resource has permanently moved and browsers should update bookmarks?

Answer: a) 301

Justification: HTTP **301 Moved Permanently** indicates that the requested resource has permanently moved to a new URL. Clients and search engines can update stored references

and bookmarks accordingly.

301: Permanent redirect

302: Temporary redirect

304: Not Modified

307: Temporary redirect while preserving the request method

14. In HTTP caching, the Cache-Control: no-cache directive means:

Answer: b) Validate with the server before using a cached copy

Justification: no-cache does **not** mean that the resource cannot be stored in a cache.

Instead, it means that the cached response must be revalidated with the origin server before being reused. This allows the client to determine whether the cached representation is still valid.

15. Which DNS record type maps a domain name to an IPv6 address?

Answer: c) AAAA

Justification: An **AAAA record** maps a domain name to an IPv6 address.

For comparison:

A: Maps a domain name to an IPv4 address.

AAAA: Maps a domain name to an IPv6 address.

CNAME: Creates an alias for another domain name.

MX: Identifies mail servers for a domain.

16. A DNS resolver switches from UDP to TCP when:

Answer: b) The response is truncated (TC bit set)

Justification: If a DNS response cannot fit within the permitted UDP response size, the

server may set the **Truncated (TC)** bit. The resolver can then retry the query using TCP to obtain the complete response. This mechanism is particularly relevant for larger DNS responses.

17. Which attack exploits the UDP-based DNS protocol by sending forged responses with a matching transaction ID?

Answer: b) DNS cache poisoning

Justification: **DNS cache poisoning** occurs when an attacker attempts to insert fraudulent DNS information into a resolver's cache. In traditional DNS, attackers could attempt to forge responses containing the correct transaction identifier and other expected information. DNSSEC and modern resolver protections help defend against such attacks.

18. Setting a very low TTL (e.g., 30 s) for a DNS record primarily increases:

Answer: b) Recursive resolver query load

Justification: TTL determines how long a DNS response can remain cached. With a TTL of only 30 seconds, cached records expire frequently. Recursive resolvers therefore need to query authoritative DNS servers more often, increasing DNS query traffic and server load.

19. A user browses to a new website. The correct sequence of protocol interactions for the first byte of the page is:

Answer: b) DNS → TCP handshake → HTTP GET

Justification: For a traditional HTTP connection, the client first needs to resolve the domain name through DNS. After obtaining the server's IP address, it establishes a TCP connection using the three-way handshake. The client can then send the HTTP GET request and receive the response.

Sequence:

DNS resolution → TCP connection establishment → HTTP request → HTTP response

Note: Modern HTTPS/HTTP/3 can use different mechanisms, but among the given options, **(b)** is the correct answer.

20. Under 20% packet loss, which protocol combination gives the best performance for a real-time multiplayer game?

Answer: b) UDP with application-layer sequencing for game state, HTTP/3 for assets

Justification: Real-time multiplayer games require low latency and can tolerate some packet loss. UDP is suitable for game-state information because it avoids the retransmission delays and strict in-order delivery of TCP. Application-layer sequencing allows the game to identify outdated or missing updates and decide how to handle them. For downloading assets, **HTTP/3**, which uses QUIC over UDP, provides reliable and secure transfer while avoiding some limitations of TCP-based HTTP. Therefore, the combination provides an appropriate balance between low-latency game-state communication and reliable asset delivery.

Question No.	Answer Option	Correct Answer
1	c	Fast Recovery
2	b	16 MSS
3	c	Window size
4	b	Buffers small segments until ACK arrives
5	b	CWND growth rate relative to RTT

6	b	DNS responses are small and connection overhead is unnecessary
8	b	1472 bytes
9	b	0-RTT or 1-RTT connection setup
10	b	Reusing the same TCP connection for multiple requests
11	b	A response blocks subsequent responses on the same connection
12	c	TCP layer
13	a	301
14	b	Validate with the server before using a cached copy
15	c	AAAA
16	b	The response is truncated (TC bit set)
17	b	DNS cache poisoning
18	b	Recursive resolver query load
19	b	DNS → TCP handshake → HTTP GET
20	b	UDP with application-layer sequencing for game state, HTTP/3 for assets

